Covert Change-Point Detection with Resource-Constrained Multi-Sensor Sampling

Your Name(s)
Affiliation
Email

***Abstract*—We study the problem of quickest change-point detection in multi-sensor systems under resource constraints and adversarial conditions. Sensors collect observations whose distribution changes at an unknown time. Unlike classical formulations, we consider a covert adversary that selects post-change distributions to remain undetected while maximizing detection delay. In addition, sensing and communication incur costs, limiting the number of sensors that can be sampled over time. We analyze the trade-offs between average detection delay, false alarm rate, sampling strategy, and communication cost. For Gaussian models, we characterize the impact of sensor selection and correlation structure on detection performance. Our results show that optimal strategies prioritize sensors based on information-per-cost ratios and that covert adversaries fundamentally alter classical detection scaling laws.**

I. Introduction

Sequential change-point detection is a fundamental problem in signal processing, networking, and cybersecurity. A decision-maker observes a stochastic process whose distribution changes at an unknown time and aims to detect this change as quickly as possible while controlling false alarms [1]–[4].

Classical formulations assume that the post-change distribution is fixed. However, in adversarial environments, attackers may adapt their behavior to avoid detection. This motivates the study of *covert change-point detection*, where the adversary selects post-change distributions to minimize detectability.

At the same time, practical sensing systems are constrained by limited resources, including energy, bandwidth, and communication costs. These constraints restrict the number of sensors that can be sampled or whose data can be transmitted.

In this paper, we study change-point detection under:

- multiple sensors,
- sampling and communication constraints,
- a covert adversary aware of the detection strategy.

Our goal is to characterize detection performance and derive principled sensor selection strategies under these constraints.

II. System Model

A. Observations

Consider M sensors indexed by $i = 1, \dots, M$. Each sensor produces a sequence $\{X_i(t)\}_{t \geq 1}$.

Before the change point τ:

$$X_i(t) \sim \mathcal{N}(\mu_{0,i}, \sigma_i^2). \tag{1}$$

After the change:

$$X_i(t) \sim \mathcal{N}(\mu_{1,i}, \sigma_i^2), \tag{2}$$

where $\mu_{1,i}$ is chosen by an adversary.

B. Correlation Structure

Let $\boldsymbol{X}(t) = (X_1(t), \dots, X_M(t))$. We allow correlated observations:

$$\boldsymbol{X}(t) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \tag{3}$$

where Σ captures sensor correlations.

C. Sampling Policy

At each time t, the detector selects a subset of sensors:

$$S(t) \subseteq \{1, \dots, M\}. \tag{4}$$

Sampling sensor i incurs cost C_i, leading to the constraint:

$$\mathbb{E}\left[\sum_{i \in S(t)} C_i\right] \leq B. \tag{5}$$

D. Communication Cost

Transmitting data from sensor i incurs cost T_i:

$$\mathbb{E}\left[\sum_{i \in S(t)} T_i\right] \leq C. \tag{6}$$

III. Detection Problem

Let ν denote the unknown change point, i.e., the time at which the statistical distribution of the observations changes. The detector observes the sequence of measurements sequentially and must decide when to declare that a change has occurred. This decision is represented by a stopping time τ, which corresponds to the detection instant.

The goal of quickest change-point detection is to detect the change as quickly as possible after it occurs while controlling the frequency of false alarms. A common performance metric is the *average detection delay* (ADD), defined as

$$ADD = \mathbb{E}[(\tau - \nu)^+], \tag{7}$$

where $(x)^+ = \max(x, 0)$. This quantity measures the expected delay between the change point ν and the detection time τ, counting only delays that occur after the change.

To prevent the detector from raising alarms too frequently, a constraint is imposed on the false alarm rate. Specifically, we require

$$\mathbb{E}_\infty[\tau] \geq \gamma. \tag{8}$$

Here, the expectation $\mathbb{E}_\infty[\cdot]$ denotes expectation under the *no-change* regime, that is, when the change never occurs (formally $\nu = \infty$). Under this regime all observations follow the pre-change distribution for all time. Therefore, $\mathbb{E}_\infty[\tau]$ represents the *average run length to false alarm*, i.e., the expected time until a false alarm is raised when the system remains in the nominal state.

The parameter γ controls the false alarm rate: larger values of γ correspond to more conservative detectors that raise alarms less frequently but typically incur larger detection delays.

IV. Detection Statistic

We extend the CuSum procedure to the multi-sensor setting. Let $\mathcal{S}(t) \subseteq \{1, \dots, M\}$ denote the subset of sensors sampled at time t. The cumulative log-likelihood statistic is

$$S_t = \max_{1 \leq k \leq t} \sum_{j=k}^{t} \sum_{i \in \mathcal{S}(j)} \log \frac{f_{1,i}(X_i(j))}{f_{0,i}(X_i(j))}. \tag{9}$$

The stopping rule declares a change when the statistic exceeds a threshold:

$$\tau = \inf\{t \geq 1 : S_t \geq h\}, \tag{10}$$

where h is chosen to satisfy the false alarm constraint $\mathbb{E}_\infty[\tau] \geq \gamma$.

V. Adversarial Model

We model the interaction between the detector and the adversary as a Stackelberg game.

- The detector selects a sampling policy $\mathcal{S}(t)$ and a detection threshold h.
- The adversary selects the post-change parameters $\mu_{1,i}$ to maximize the detection delay.

The adversary is assumed to be *covert*, meaning that the post-change distribution approaches the pre-change distribution as the false alarm constraint becomes more stringent. Formally,

$$\mu_{1,i}(\gamma) \to \mu_{0,i} \quad \text{as } \gamma \to \infty. \tag{11}$$

VI. Performance Analysis

For Gaussian observations, the Kullback–Leibler divergence between the pre-change and post-change distributions at sensor i is

$$D_i = \frac{(\mu_{1,i} - \mu_{0,i})^2}{2\sigma_i^2}. \tag{12}$$

If a subset $\mathcal{S}$ of sensors is used, the effective divergence is

$$D_{\text{eff}} = \sum_{i \in \mathcal{S}} D_i. \tag{13}$$

In classical quickest change detection problems, the average detection delay satisfies

$$ADD \sim \frac{\log \gamma}{D_{\text{eff}}}. \tag{14}$$

However, under covert adversarial behavior the divergence may vanish with γ,

$$D_{\text{eff}}(\gamma) \to 0, \tag{15}$$

which leads to fundamentally different scaling,

$$ADD = \Theta(\gamma). \tag{16}$$

VII. Optimal Sensor Selection

Sensors should be prioritized according to their information-per-cost ratio,

$$\frac{D_i}{C_i + T_i}, \tag{17}$$

where C_i represents the sensing cost and T_i represents the communication cost.

Insight:

- Sensors with higher divergence provide stronger evidence of change.
- Correlated sensors yield diminishing information gains.
- Communication cost can significantly influence the optimal sensor subset.

VIII. Continuous-Time Extension

In a continuous-time model, sensor observations follow stochastic differential equations of the form

$$dX_i(t) = \begin{cases} \mu_{0,i}\, dt + \sigma_i\, dW_i(t), & t < \nu, \\ \mu_{1,i}\, dt + \sigma_i\, dW_i(t), & t \geq \nu, \end{cases} \tag{18}$$

where $W_i(t)$ are correlated Brownian motions.

In this setting, the detection problem reduces to monitoring a continuous-time likelihood ratio process under sampling constraints.

IX. Discussion

We identify three regimes:

- Low budget: single best sensor.
- Medium budget: subset selection.
- High budget: all sensors.

Correlation reduces effective information gain, making sensor diversity critical.

X. Conclusion

We studied covert change-point detection with resource-constrained multi-sensor sampling. Our results show that adversarial behavior fundamentally changes detection scaling laws and that optimal strategies must jointly consider sensing cost, communication constraints, and statistical informativeness.

Future work includes decentralized detection, adaptive policies, and learning-based approaches.

REFERENCES

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